



Negative Feed Back (General theory Of Feed back)

Derivation of negative feedback equation:

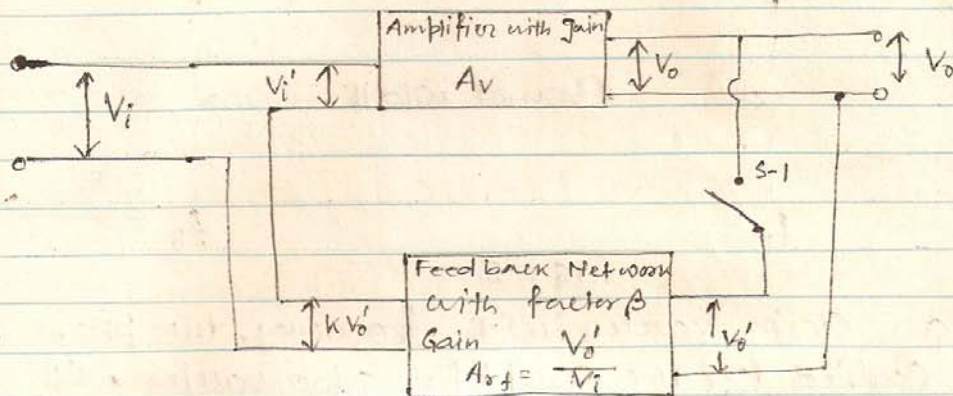


FIG:1 Block diagram of an amplifier with a feedback circuit.

In the amplifier shown in fig-1 with S-1 open the output is V_o and with S-1 closed is V_o' .

The amplifier of voltage gain A_v has a network connected between its input and output terminals, so that when S-1 is closed some of the output of the amplifier can be fed back to the input.

(i) Now consider S-1 open:

The output of the network is zero so that input to amplifier is $V_i + 0 = V_i$. Let the output of amplifier be V_o then

$$A_v = \frac{V_o}{V_i}$$

where A_v = voltage gain of the amplifier without feedback.



Feedback

i) Now S-1 is closed, a certain amount of output (V_o) will be fed back into the input in series with V_i . This will change the output. Let now the output of the amplifier with feedback be V_o' .

The amount of voltage fed back by the feedback network will be $\beta V_o'$ where

(1) $\beta < 1$ when network consists of ^{passive} positive elements.

and (2) $\beta > 1$ when network contains an active device.

Then the Network input amplifier = $(V_i - \beta V_o')$
The sign of β will depend upon the phase ^(A) relationship.

Now as,

V_o' = New output from the amplifier

We get using (A) :

$$V_o' = A_v (V_i - \beta V_o')$$

$$\text{or } V_o' (1 + \beta A_v) = A_v V_i$$

$$\text{or } \frac{V_o'}{V_i} = \frac{A_v}{1 + \beta A_v} \quad \text{--- (1)}$$

But $\frac{V_o'}{V_i} = A_{vf}$ = Gain of the amplifier with feedback.

So that eqⁿ (1) becomes:

$$A_{vf} = \frac{A_v}{1 + \beta A_v} \quad \text{--- (2)}$$

Eqⁿ (2) can also be written in the following form

$$A' = \frac{A}{1 + A\beta} \quad \text{--- (3)}$$

Eqⁿ (2) and (3) is called feedback equation.

Discussion: We have from eqⁿ (3) $1 + A\beta = \frac{A}{A'}$



Feedback

If A' is less than A ($A' < A$)

$$\text{or } \frac{A}{A'} > 1$$

$$\therefore [1 + A\beta > 1]$$

In this case β is negative, the feedback is negative and the output is reduced. Further the relation between this fraction of feedback and the reduction in gain is non-linear.

(i) When β is positive, the feedback is positive and the voltage gain is increased. Then $1 + A\beta < 1$

Why use negative feedback?

The reasons are as follows:

(a) Increase stability: In eqⁿ (2) if β is negative and $\beta A_v \gg 1$ then $A_{vf} = \frac{A_v}{1 - \beta A_v} \approx \frac{A_v}{\beta A_v} = \frac{1}{\beta}$ (4)

Thus the gain of the amplifier will depend only upon the feedback network (eqⁿ 4) and changes in parameter (such as β of a transistor) will affect the gain of the amplifier without feedback but not affect the overall gain with feedback.

(b) Increase bandwidth: The bandwidth is frequency range over which the power gain remains a constant gets increased.

(c) Decrease distortion: [When a large amplitude signal is applied to the input of an amplifier the output signal is distorted due to non-linearity in the transfer characteristics of the amplifier active device. Thus a distortion voltage in addition to the signal component appears at the output of the amplifier.]

If an amplifier has a voltage gain A_v



and the total harmonic distortion D then the application of (-)ve feedback will reduce the distortion according to the relation given below

$$D_f = \frac{D}{1 + \beta A_v} \quad \#$$

(d) Decrease noise: A similar expression as above gives the reduced noise. The relation is obtained by replacing D by N .

$$N_f = \frac{N}{1 + \beta A_v} \quad +$$

(e) Input Resistance:

An increase in input resistance of an amplifier is generally considered to be an advantage since it reduces loading on previous stage.

If the output voltage is f.b in series with the input. Then the input current decreases and thus the effective input resistance increases given by

$$R_{if} = R_i (1 + \beta A_v)$$

as $|1 + \beta A_v| > 1$

then $R_{if} > R_i$

where β has only magnitude.

(f) Output Resistance: Impedance:

If output voltage is returned to the input (either in series or shunt), the output resistance decreases (R_{of}) where as if a signal proportional to the output current is used as feedback, the output resistance increases.

Z_o is the output impedance of the amplifier without f.b

$$Z_{of} = \frac{Z_o}{1 + \beta A_v} \quad \because |1 + \beta A_v| > 1 \quad \text{then } Z_{of} < Z_o$$

N.B

+ve f.B is never applied to an amplifier because then it begins to function as an "Oscillator".



Feedback

Types of (-)ve f.b in Transistor Circuits:

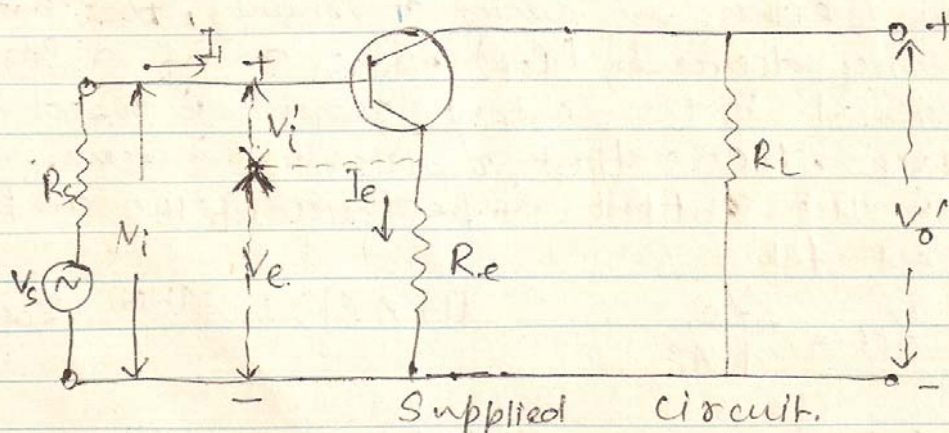
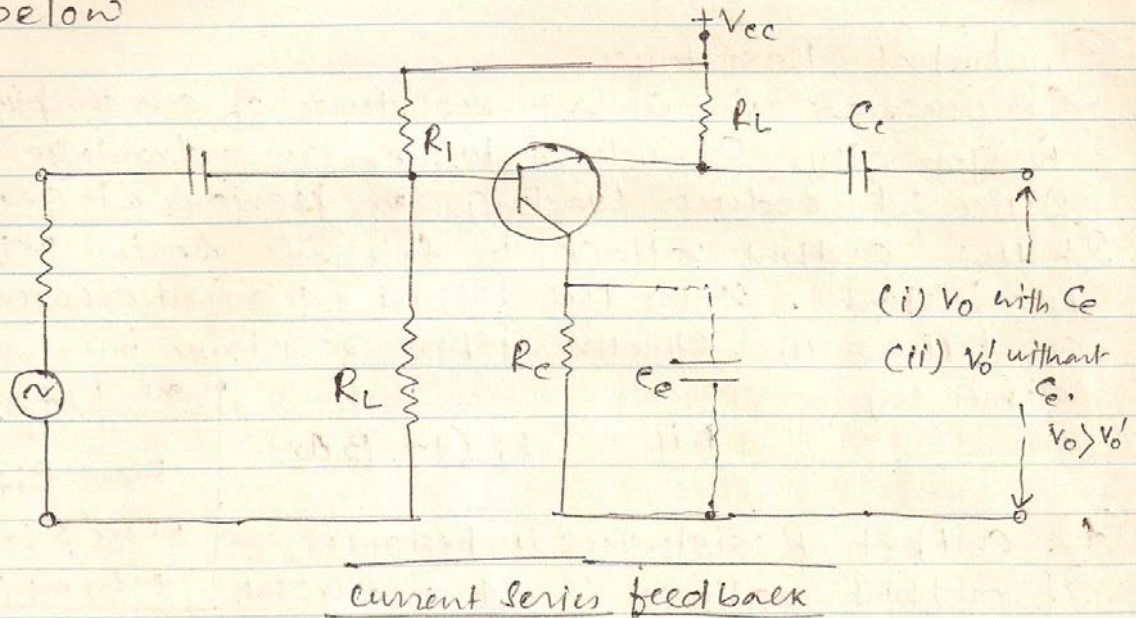
There are four f.b topologies namely

- (1) Voltage Series f.b.
- (2) Voltage-Shunt f.b.
- (3) Current-Series f.b.
- (4) Current-Shunt f.b.

The current series feedback is discussed below

The current series f.b is usually used for achieving (-)ve f.b.

A most commonly used ~~type~~ circuit is shown below





Feedback

FIG:1: A Common method of introducing current f.b is to place a resistor (R_e) in the emitter load of C-E amplifier. when R_e is bi-passed with C_e , the output is V_o and the gain without f.b is A_v . when C_e is removed an a.c voltage will be developed across R_e due to the flow of emitter current. This voltage will serve to reduce the input voltage between base and emitter so that the output voltage will drop to V_o and the gain with -ve f.b is now A_{vf} .

(a) Expression for Feedback factor K_o

FIG:2 We have:

$$\begin{aligned} V_o' &= -I_c R_L \quad \text{--- (1)} \\ \text{and } V_e &= I_e R_e \approx I_c R_e \quad \text{--- (2)} \end{aligned} \quad \left| \begin{array}{l} V_o' \text{ is} \\ \text{without } C_e \\ I_e \approx I_c \end{array} \right.$$

$$\therefore V_e = -\frac{R_e}{R_L} \cdot V_o' \quad \text{--- (3)} \quad \left| \begin{array}{l} \text{from (2)} \\ \text{(2)} \end{array} \right.$$

Thus, the fraction of output voltage f.b in series with the input is given by:

$$|K| \approx \frac{R_e}{R_L} \quad \text{--- (4)}$$

(b) Expression for Internal resistance (R_{if}) [effect of f.b on input impedance]

We have,

$$R_{if} = \frac{V_i}{I_i} = \frac{V_i' + V_e}{I_i} = \frac{V_i'}{I_i} + \frac{V_e}{I_i}$$

But $\frac{V_i'}{I_i} = R_i$; $V_e = I_e R_e$ and $I_i = I_e - I_c$.

$$\begin{aligned} \therefore R_{if} &= \left(R_i + \frac{I_e R_e}{I_e - I_c} \right) \\ &= \left\{ R_i + \frac{R_e}{\left(1 - \frac{I_c}{I_e} \right)} \right\} \quad \text{--- (5)} \end{aligned} \quad \left| \begin{array}{l} \text{Circuit diagram showing } I_i \text{ entering the base, } I_c \text{ leaving the collector, and } I_e \text{ leaving the emitter.} \end{array} \right.$$



Feedback

16:1 A Common method of introducing current f.b is to place a resistor (R_e) in the emitter load of C-E amplifier. when R_e is bi-passed with C_e , the output is V_o and the gain without f.b is A_v . when C_e is removed an a.c voltage will be developed across R_e due to the flow of emitter current. This voltage will serve to reduce the input voltage between base and emitter so that the output voltage will drop to V_o and the gain with -ve f.b is now A_{vf} .

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V_o' is without C_e
 $I_e = I_c$
 from (2):

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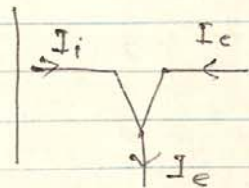
(b) Expression for Internal resistance (R_{if}) [effect of f.b on input impedance]

We have,

$$R_{if} = \frac{V_i}{I_i} = \frac{V_i' + V_e}{I_i} = \frac{V_i'}{I_i} + \frac{V_e}{I_i}$$

But $\frac{V_i'}{I_i} = R_i$; $V_e = I_e R_e$ and $I_i = I_e - I_c$.

$$\begin{aligned} \therefore R_{if} &= \left(R_i + \frac{I_e R_e}{I_e - I_c} \right) \\ &= \left\{ R_i + \frac{R_e}{\left(1 - \frac{I_c}{I_e}\right)} \right\} \quad \text{--- (5)} \end{aligned}$$





Again, $\frac{I_c}{I_e} = h_{fb} = \frac{h_{fe}}{1 + h_{fe}} \quad \text{--- (6)}$

So that from (5) & (6) we have

$$R_{if} = R_i + \frac{R_e}{1 - \frac{h_{fe}}{1 + h_{fe}}} = R_i + \frac{R_e(1 + h_{fe})}{1}$$

Thus,

$$R_{if} = R_i + (1 + h_{fe}) R_e \quad \text{--- (7)}$$

But,

$$R_i = h_{ie}$$

So that eqⁿ (7) becomes: $R_{if} = h_{ie} + (1 + h_{fe}) R_e$

(c) voltage gain (A_{vf}):

Since,

$$A_v = \frac{A_i R_L}{R_i}$$

$$\therefore A_{vf} = \frac{A_i R_L}{R_{if}} \quad \text{--- (9)}$$

Now, $A_i \approx -h_{fe}$

So that we obtained $A_{vf} = - \frac{h_{fe} R_L}{R_{if}} = - \frac{h_{fe} R_L}{h_{ie} + R_e}$

(d) Output Resistance: (R_{of}):

We have,

$$R_{of} \approx \frac{1 + h_{fe}}{h_{oe}} \approx \frac{1}{h_{oe}} \quad \text{--- (11)}$$

which is a very large value in the Megohm range, so that the overall output resistance R_{of} in including the load resistance is approximately R_L .



balance, the voltages across the lower bridge and V_1 and V_2 are equal and given by

$$V_1 = V_2 = \left(\frac{V_0}{R_1 + R_2} \right) R_2 \quad \left| \quad V = (i) R_2 \right.$$

$$= \frac{V_0}{2 R_2 + R_2} \cdot R_2 = \frac{1}{3} V_0$$

Thus at balance we have ; $V_1 = V_2 = \frac{V_0}{3}$

$$\text{and } \frac{R_2}{R_1 + R_2} = \frac{1}{3}$$

$$\text{or } \frac{V_1}{V_0} = \frac{1}{3} = \frac{V_2}{V_0}$$

As pointed out earlier, the output voltage V_i from the bridge must not be zero for oscillations to occur. This is accomplished by taking the ratio $\frac{R_2}{R_1 + R_2}$ smaller than $\frac{1}{3}$.

$$\text{Thus we let, } \frac{R_2}{R_1 + R_2} = \frac{1}{3} - \frac{1}{K}$$

where $K > 3$, the bridge is then slightly unbalanced and given a feedback voltage V_i .

Since the bridge is made unbalanced by changing the ratio $\frac{R_1}{R_2}$, the ratio $\frac{V_1}{V_0}$ remains $\frac{1}{3}$ but the ratio $\frac{V_2}{V_0}$ is now equal to $\frac{1}{3} - \frac{1}{K}$.

The voltage V_i is then

$$V_i = V_1 - V_2 = \left(\frac{1}{3} - \frac{1}{3} + \frac{1}{K} \right) V_0 = \frac{V_0}{K}$$

Thus, V_i is in phase with V_0 and the reverse transmission factor β is

$$\beta = \frac{V_i}{V_0} = \frac{1}{K}$$

Clearly the condition $A\beta = 1$ is satisfied if the gain of the oscillator amplifier is made equal to K which is greater than 3.



Wien-Bridge Oscillator:

A two stage CE-transistor amplifier may be used for this purpose.

The Wein-bridge Oscillator is an RC Oscillator in which a (nearly) balanced bridge is used as the feedback network. This bridge introduces no phase-shift between its input and output voltages. The amplifier should also not introduce any phase shift so that the net phase shift around the loop is zero.

The condition for V_2 to be in phase with V_1 are met with when $X_C = R$. Under this condition ($V_2 = \frac{V_1}{3}$), the attenuation is not as high as in phase shift network.

Variable frequency operation requires the sinusoidal adjustment of the two capacitors, an easier task than in the phase shift.

This bridge oscillator is capable of generating frequency from 20 Hz to 200 kHz with a distortion of less than 1%.

This is the oscillator which can generate readily a variable frequency output. Like a phase shift oscillator, a Wein-Bridge Oscillator is ordinarily used to generate A.F signals.

1988

Qr. Compare the characteristic phase-shift and Wien-Bridge Oscillator.

Phase Shift Oscillator: The transistor Phase Shift Oscillator operates in the CE mode and so it produces a phase shift of 180° between the input and output voltages.



An additional 180° shift of phase is produced by the three sections of R-C Network. Hence the output voltage after three sections which is feed back to the input of the amplifier would be in phase with input signal.

By changing the value of R & C simultaneously in each section, the frequency of oscillator can be changed from as low as 20 Hz to as much as 300 kHz .

It is to be noted that there is only one frequency at which the net phase shift around the loop is 0° or 360° so that a sine wave output is obtained.

The phase shift are commonly used for audio frequency ranges.

CRYSTAL OSCILLATOR (VHF and UHF)

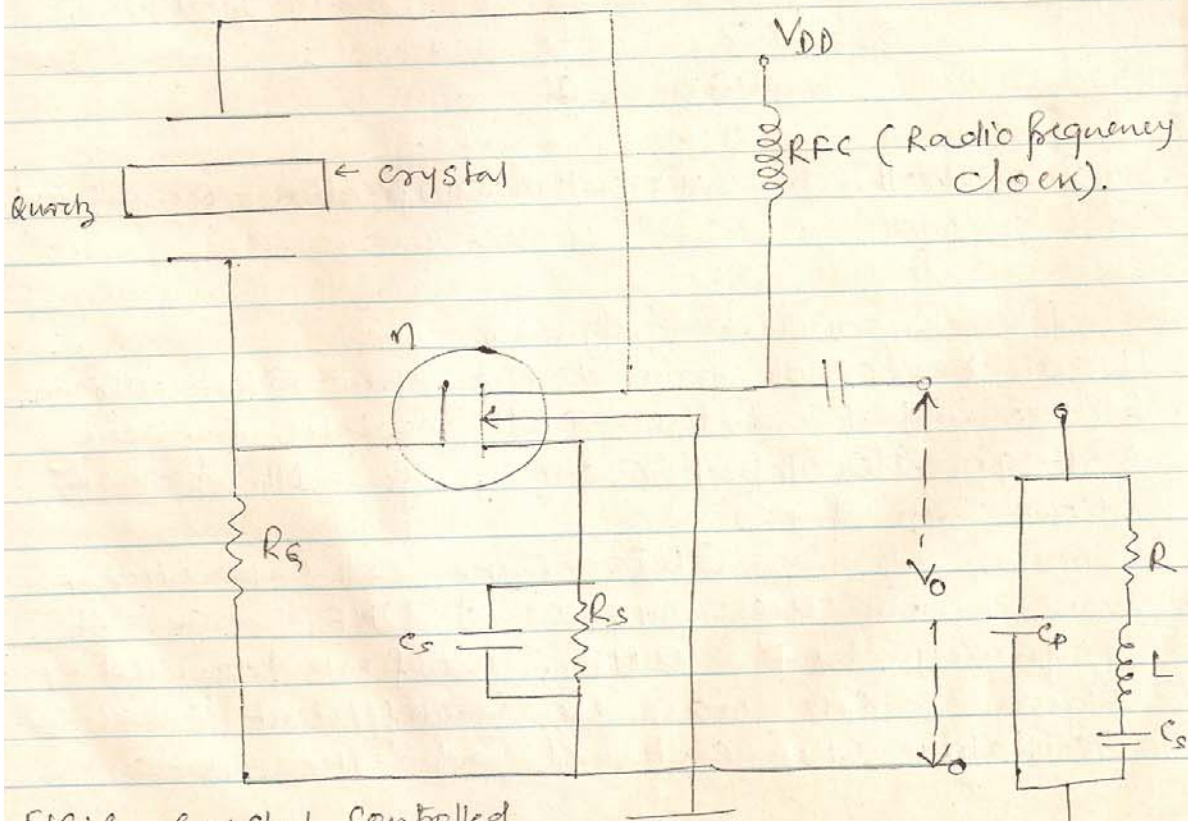
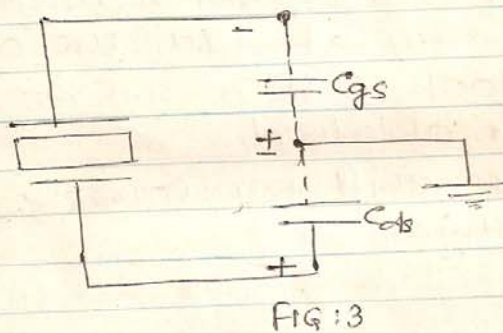


FIG:1 Crystal Controlled Oscillator using an N-channel MOSFET (VHF & UHF range)

FIG:2 Equivalent circuit



The Crystal Controlled Oscillator is shown in the feedback is achieved through the interelectrode capacitances (C_{gs} and C_{ds} where C_{gs} = gate to source capacitance and C_{ds} = drain to source capacitance). These capacitors provide a tapped capacitor across the crystal which acts like a high Q parallel resonant circuit. (as shown in fig: 3 separately tuned to some accurately known frequency.



Since resistance R of the crystal is very small, the Q of the crystal is very high. Hence the frequency stability of the crystal is also very high.

Now,

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f}$$

$$Q = \text{voltage current magnification} = \text{quality factor} \\ = \frac{\omega_0 L}{R}$$

f_0 = Resonant frequency.
Phase reversed from the output at the drain to the input at the gate results from the voltage developed across C_{gs} with polarity shown in fig: 3

This type of capacitor is useful in the VHF and UHF ranges. At frequency below 2 MHz, the $f-b$ provided by C_{gs} and C_{ds} may be insufficient to maintain oscillation when external capacitor will have



be added to supplement the Transistor Capacitance

Discussions on frequency Stability of Oscillators:

frequency Stability of Oscillators:

For most purpose it is of importance that the operative frequency of an oscillator must remain constant. But frequency of an oscillator changes due to a number of factors. The generated frequency is given by an expression of the form:

$$f = f_0 (1 + k)$$

where,

f_0 = resultant frequency of the tune circuit.
and k = constant, much less than unity ($k \ll 1$)
usually k is directly proportional to the loss-resistance ($L = 0$, ohmic resistance) of the tune circuit and inversely proportional to the f_0 resistance (r_e) of the transistor. The coupling of the resistance load to the tune circuit increases the loss resistance. Thus the variation in load causes a change in the value of k . Similarly a change in supply voltage also causes a variation in the value of k . Hence the change in frequency will occur due to the following factors:

(i) variation of supply voltages:

Any change in the operating voltage applied changes the internal resistance (r) of the active transistor device and hence the frequency of oscillator. This type of frequency change is called the "Dynamic instability" and can be overcome by using an electronically regulated power supply.



(ii) Variation in Load:

A change in the load on device will bring a change in collector voltage and hence a shift in the frequency of the oscillator. Its effect can be reduced by interposing an amplifier stage between the oscillator and power consuming load. An amplifier used in this manner to isolate an oscillator from its load is known as "buffer amplifier". Such an amplifier is necessary when the frequency generated is required to have a high degree of stability.

(iii) Variation in the reactive components of the tuned circuit due to change in temp:

Temperature changes affects slightly the inductance and capacitance of the tuned circuit causing a change in the resonant frequency (f_0) of the oscillator. This effect can be minimised by the use of (a) Adequate Ventilation & (b) a low direct input voltage.

(iv) Variation in the inter electrode:

Capacities due to Thermal changes:

A change in temperature causes the inter electrode capacitance to vary. Now, since these capacitance form a part of the tuned circuit the frequency of the oscillator will be affected. This effect may be minimised by shunting them by a large valued capacitance.

(v) Mechanical vibrations:

Mechanical vibration of the circuit elements such as inductors and capacitors cause these values to change and thus change the resonant frequency. This variation may be minimised by mounting the elements rigidly.



insulating them from the oscillator.

) Ageing of the components: The parameters of the active device change due to ageing and hence the frequency generated is affected.

Amplitude Control:

In practical oscillators, the circuit is so designed that

$$|A\beta| > 1$$

and that the loop gain $(1+A\beta)$ never falls below unity due to parameter variations.

As the oscillations build up in amplitude, the non linearity of the device limits the amplitude. In all the resonant circuit oscillators the biasing circuit used develops a larger negative bias with the build up of oscillation. The average transconductance of the device goes on decreasing as the signal swing to the input of the amplifying device, increases. The steady state amplitude is reached when g_m decreases sufficiently to make,

$$|A\beta| = 1$$

If there is any further amplitude variation the negative feedback provided by the biasing circuit tends to stabilise the amplitude.